

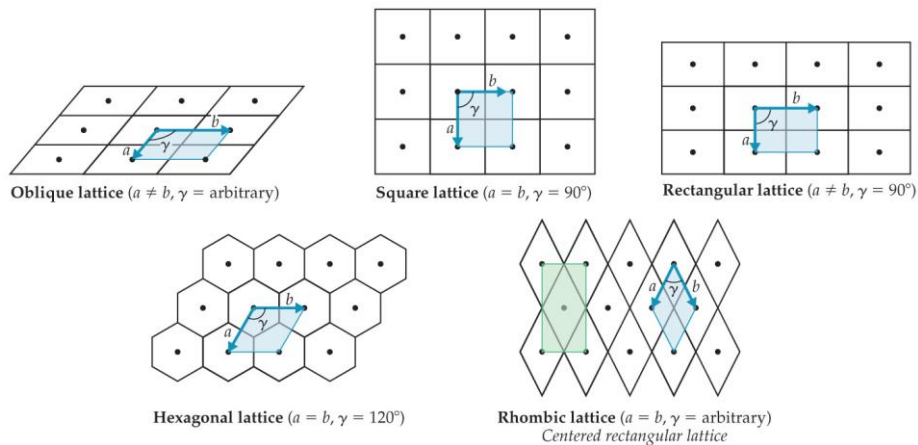
Chapter 12

Solids and modern materials

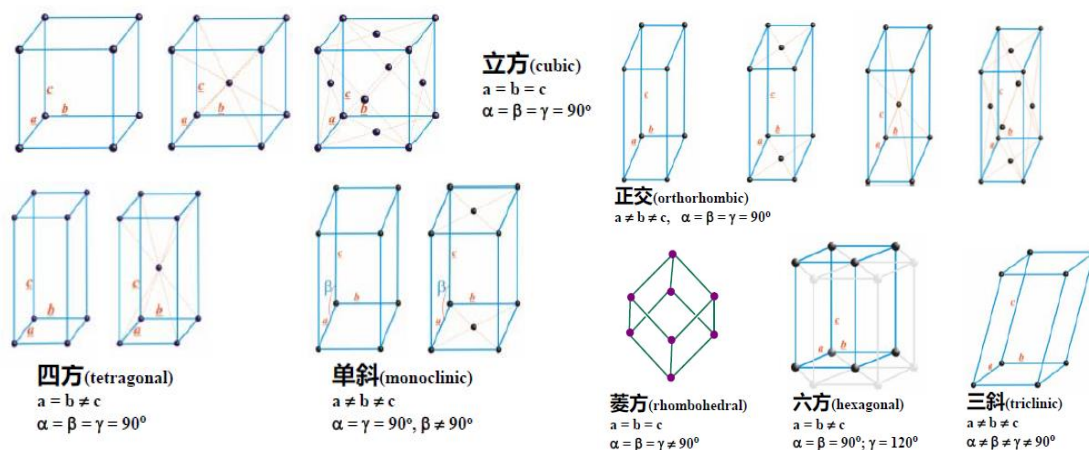
Review

Definitions:

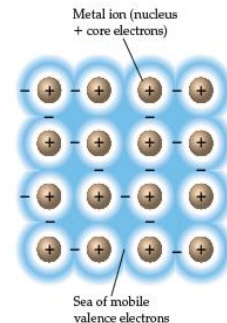
- **Organization of solids:**
 - 1) **Crystalline solids:** the atoms are arranged **in an orderly repeating pattern.**
 - 2) **Amorphous solids:** **lack of order** in the arrangement of atoms.
- **Comparison:**
 - 1) Crystalline solids have regular geometric shape and fixed melting point, while amorphous solids do not.
 - 2) The properties of electricity, mechanics, optics, thermal conductivity, electrical conductivity **for crystalline solids are anisotropy, while amorphous solids is isotropy.**
- **Unit cell:** a relatively small repeating unit in a crystalline solid.
- **Crystal lattice:** the geometrical pattern of points on which the unit cells are arranged.
- **Lattice point:** positions that define the overall structure of the crystalline compound.
- **Lattice vector:** the positions of the lattice points.
- **Two-dimensional lattices:**



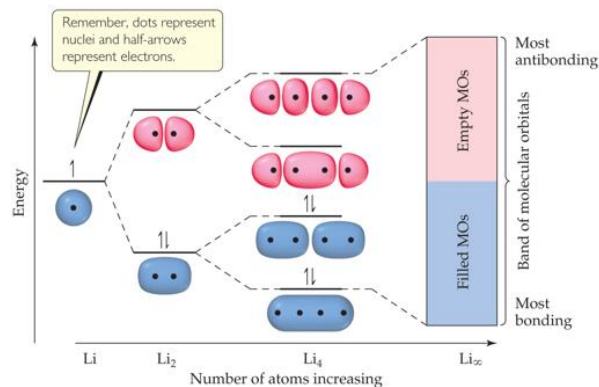
● Three-dimensional primitive lattices:



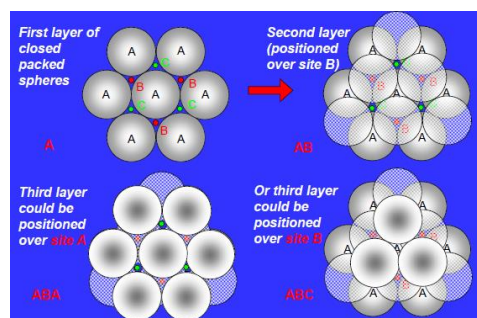
- **Classification of solids:**
 - 1) Metallic solids
 - 2) Ionic solids
 - 3) Covalent-network solids
 - 4) Molecular solids
 - 5) Polymers
 - 6) Nanomaterials
- **Metallic Solids:** held together by a **delocalized “sea” of collectively shared valence electrons.**
- **Electron-Sea model:** a simple model that pictures the metal as an array of metal cations in a “sea” of valence electrons. The electrons are confined to the metal by electrostatic attractions to the cations, and they are uniformly distributed throughout the structure.



- **Molecular-Orbital model:** **as the number of atoms in a chain increases, the energy gap between the bonding orbitals and between the antibonding orbitals disappears, resulting in a continuous band of energy.**

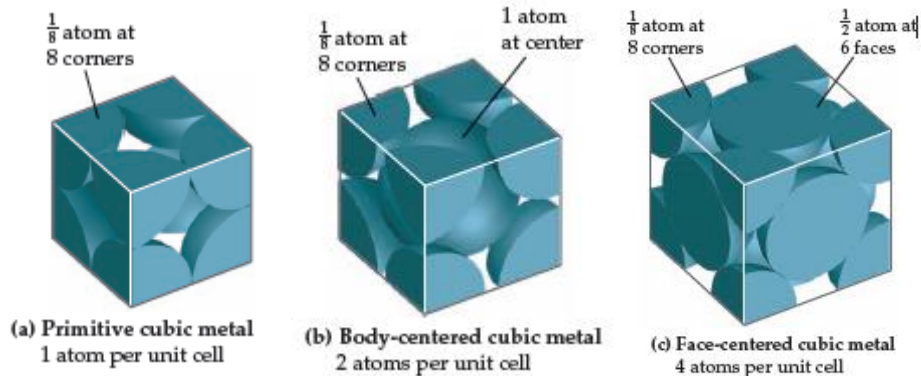
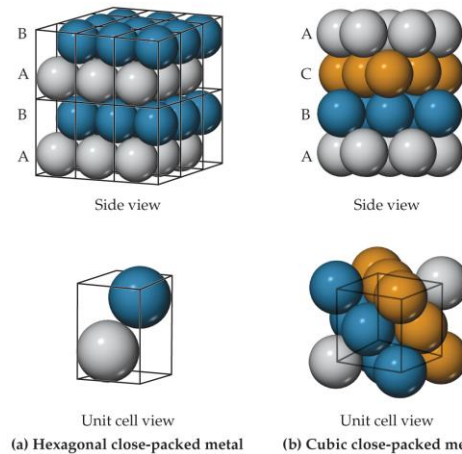


- **Band:** as the chain length goes to infinity, the allowed energy states become a continuous band.
- **Band structure:** the electronic structure of a bulk solid.
- **Close packing:**

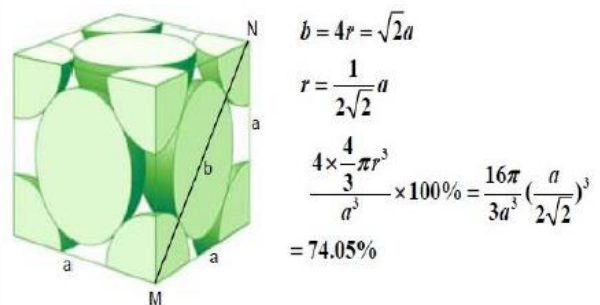
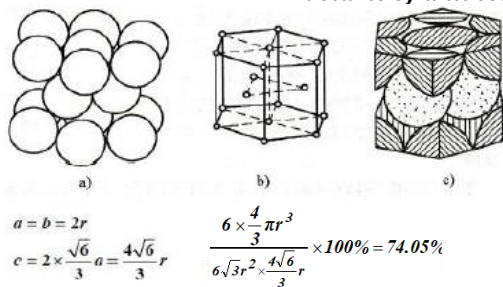


Hexagonal close packing (hcp)

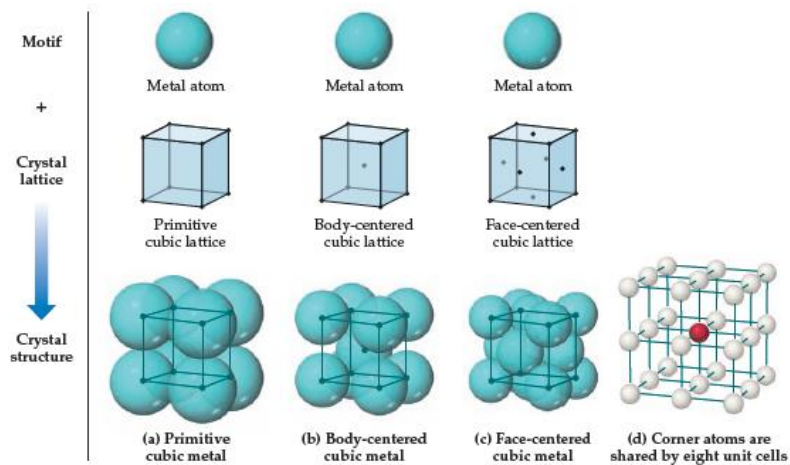
Cubic close packing (ccp)



- **Packing efficiency:** $= \frac{\text{volume of atoms}}{\text{volume of unit cell}}$



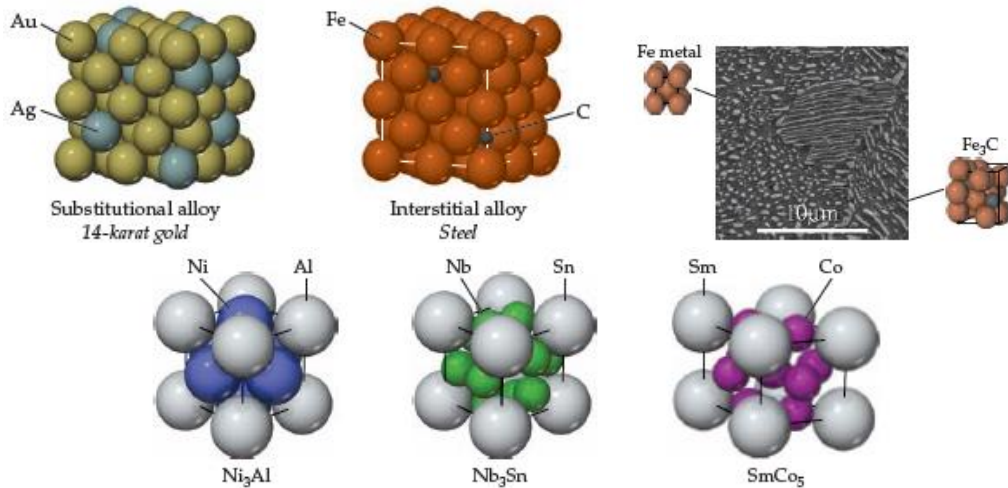
- **Structures of metallic solid:**



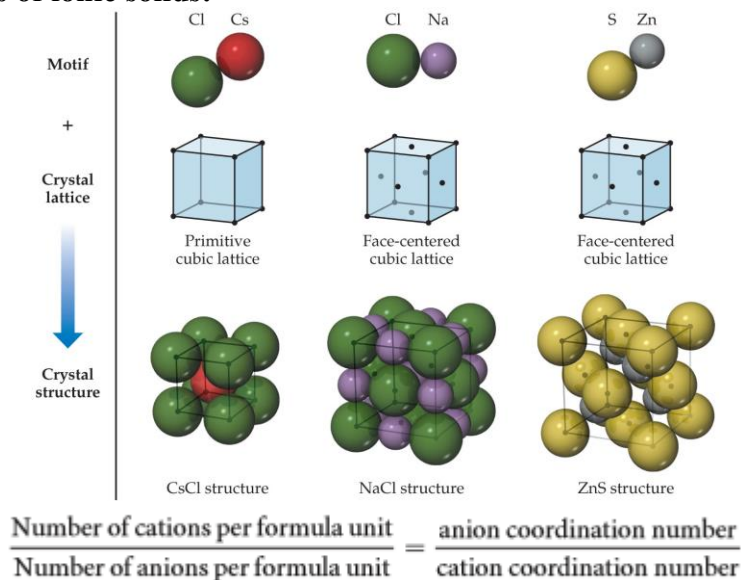
- **Alloy:** a material that contains more than one element and has the characteristic properties of a metal.
- 1) **Substitutional alloy:** Atoms of the solute in a solid solution occupy positions

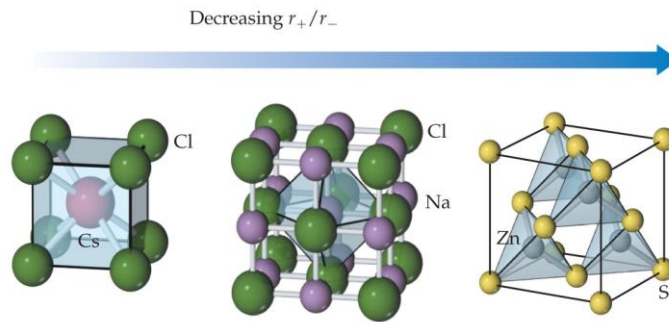
normally occupied by a solvent atom. **Two metallic components have similar atomic radii and chemical-bonding characteristics**

- 2) **Interstitial alloy:** the solute atoms occupy interstitial positions in the “holes” between solvent atoms. The solute atoms must have a much smaller bonding atomic radius than the solvent atoms.
- 3) **Heterogeneous alloy:** the components are not dispersed uniformly.
- 4) **Intermetallic compound:** compounds rather than mixtures. have definite properties and their composition cannot be varied.



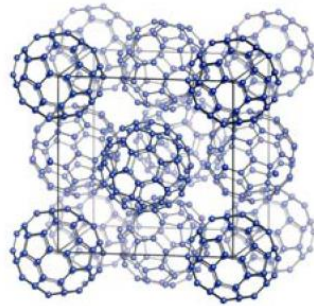
- **Ionic solids:** the substances that held together by the electrostatic attraction between cations and anions: ionic bonds.
- For a given ionic compound, a number of factors that come into play when considering which type of structure is most favorable, but two of the most important are the **relative sizes of ions** and the **stoichiometry**.
- **Structures of ionic solids:**



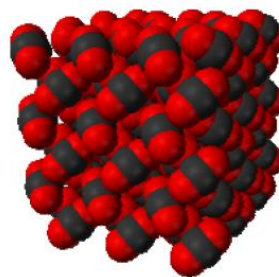


	CsCl	NaCl	ZnS
Cation radius, r_+ (Å)	1.81	1.16	0.88
Anion radius, r_- (Å)	1.67	1.67	1.70
r_+/r_-	1.08	0.69	0.52
Cation coordination number	8	6	4
Anion coordination number	8	6	4

- These structures (CsCl, NaCl, ZnS) can be realized only for ionic compounds in which the number of cations and anions is equal.
- **Molecular solids:** the substances that consist of atoms or neutral molecules held together by dipole-dipole forces, dispersion forces, and/or hydrogen bonds.

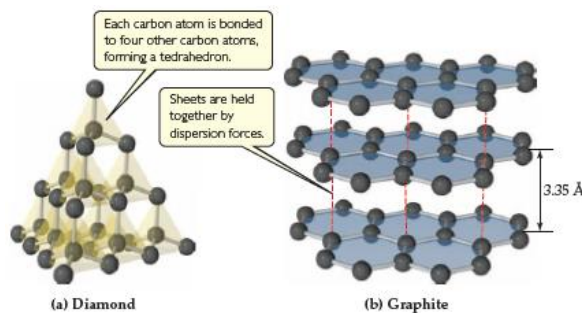


Fullerene

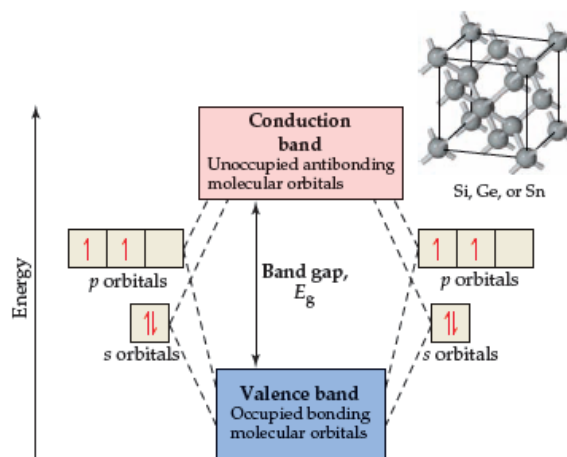


Dry ice

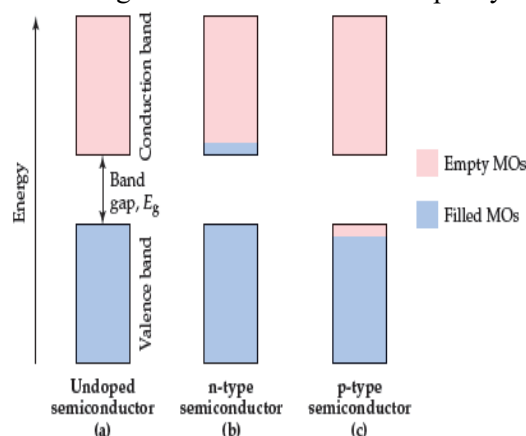
- **Covalent-network solids:** the substances that consist of atoms held together in large networks by covalent bonds.



- **Semiconductors:** solids conduct electricity but nowhere near as well as metals.
 - 1) **elemental semiconductors:** contain only one type of atom.
 - 2) **compound semiconductors:** contain two or more elements.



- **Valence band:** The band that forms from bonding molecular orbitals.
- **Conduction band:** the band that forms the antibonding orbitals.
- **Band gap (E_g):** the energy difference between the top of the valence band and the bottom of the conduction band in insulators and semiconductors.
- ✓ In a semiconductor, the valence band is filled with electrons and the conduction band is empty. These two bands are separated by the energy band gap.
- ✓ The size of the band gap increases as the bond length decreases, and as the difference in electronegativity between the two elements increases.
- **Insulator:** solids does not conduct electricity at all.
- **Doping:** The process of adding controlled amounts of impurity atoms to a material.



- **n-type semiconductors:** have more electrons, so the negative charge travels in the conduction band.
- **p-type semiconductors:** have fewer electrons, so the “hole” travels in the valence band.

Comparison:

Metallic solids	Ionic solids	Molecular solids	Covalent-network solids
✓ Consists of metal atoms held together by metallic bonding; ✓ high electrical and thermal conductivity; ✓ e.g. Silver, copper, alloys, etc.	✓ Consists of cations and anions held together by electrostatic attraction (ionic bonds); ✓ high melting and boiling points ✓ e.g. NaCl, CsCl, ZnS, etc	✓ Consists of atoms or neutral molecules held together by intermolecular forces (dipole-dipole, dispersion forces, and/or hydrogen bonds); ✓ Soft, relatively low melting point ✓ e.g. Ar, H₂O, CO₂, etc	✓ Consists of atoms held together in large networks by covalent bonds; ✓ Much harder, higher melting points than molecular solids ✓ e.g. diamond, graphite, etc

- **Polymers:** contain long chains of atoms (usually C), where the atoms within a given chain are connected by covalent bonds and adjacent chains are held to one another largely by weaker intermolecular forces.
- ✓ **Addition polymerization:** monomers are coupled through their multiple bonds.
- ✓ **Condensation polymerization:** two molecules are joined to form a larger molecule by elimination of a small molecule, such as H_2O .
- **Nanomaterials:** materials of which a single unit is sized in at least one dimension) between 1~1000 nm, but usually is 1~100 nm.